Technical Write-Up - GoComet Nova Trade Document Pipeline

Architecture

```mermaid
flowchart LR
    User["User / CG Operator"] --> UI["Minimal UI"]
    UI --> API["FastAPI Backend"]
    API --> ORCH["Pipeline Orchestrator"]
    ORCH --> EXT["Extractor Agent"]
    EXT --> VAL["Validator Agent"]
    VAL --> ROUTE["Router Agent"]
    EXT --> DB["SQLite pipeline_runs"]
    VAL --> DB
    ROUTE --> DB
    UI --> QUERY["Query Endpoint"]
    QUERY --> DB
```

The pipeline is intentionally simple. FastAPI accepts a file upload, creates a `document_id`, and stores a run record in SQLite. The orchestrator ther

Three Nastiest Failure Modes

1. Plausible but wrong extraction with high confidence

During testing, the noisy sample image made field boundaries less clear than the clean invoice. A production vision model could easily read a similar

2. Missing required field incorrectly treated as a mismatch

The more dangerous path is when a required field is absent or unreadable and the system acts as though it confidently knows the value is wrong. Th

3. Router overcommits when the signal is weak

If several required fields are uncertain, a free-form router prompt can still force an overconfident decision. The mitigation in this POC is deterministic

Observability

Every run is keyed by `document_id`, which should appear in logs, traces, and any downstream notifications. In production I would log agent name,

Cost and Latency

The slowest and most expensive step is extraction because vision understanding dominates both token usage and response time. A reasonable bac

What I'd Change With a Week

With more time I would add email ingestion, a better eval harness with labeled trade documents, and